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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I'll start by examining the contents of the dataset to understand its structure and key characteristics. Then, I'll be ready to answer any specific analytical questions you have.

Let me first load and inspect the dataset.

The dataset contains 231,636 rows and 19 columns with no missing values. Key fields include:

- **StateCode, MunicipalityCode:** Identifiers for states and municipalities.
- **MonthIssued:** The month when the contract was issued.
- **ProgramCode, SubProgramCode:** Classification codes for programs.
- **FundingValue, InvestmentValue, CommercializationValue, IndustrializationValue:** Financial values allocated for different activities.
- **ContractTotalValue:** The total value of each contract.

Now, what specific analytical question would you like me to address using this dataset? 🚀 🔄

What are the variables that most influence the differentiation between institutions that contribute the most and the least to the select the best one.

To determine the key variables that differentiate financial institutions that contribute the most vs. the least to the **ContractTotalValue**, we can take three analytical approaches:

Option 1: Descriptive Statistics & Ranking

- **Method:**
 - Aggregate **ContractTotalValue** per **FinancialInstitutionCNPJ**.
 - Rank institutions based on total contributions.
 - Compare the top 10% vs. bottom 10% institutions in terms of their average values for other financial variables (e.g., **FundingValue**, **InvestmentValue**, **CommercializationValue**, **IndustrializationValue**).
- **Pros:** Simple and interpretable.
- **Cons:** Doesn't account for complex variable interactions.

Option 2: Correlation & Feature Importance (Regression/Tree-Based Model)

- **Method:**
 - Train a regression model (e.g., linear regression, decision tree) to predict **ContractTotalValue** based on other numerical features.
 - Identify which variables have the strongest influence on contract values.
- **Pros:** More detailed insight into variable importance.
- **Cons:** Requires careful handling of potential multicollinearity.

Option 3: Clustering & Segmentation

- **Method:**
 - Use **unsupervised learning (K-Means, Hierarchical Clustering)** to group financial institutions based on contract value and related financial metrics.
 - Identify which variables most separate high-contributing vs. low-contributing institutions.

- **Pros:** Uncovers natural groupings in data.
- **Cons:** Requires choosing an appropriate number of clusters.

Best Option: Option 2 (Regression & Feature Importance)

This method not only identifies which variables most influence contract values but also quantifies their impact, making it the most actionable choice for decision-making.



Key Findings

Using a **Random Forest Regressor**, we identified the most influential variables in differentiating financial institutions based on their **ContractTotalValue**:

1. **FundingValue** (36.96% importance) – The strongest predictor. Institutions providing more funding contribute more to total contract values.
2. **InvestmentValue** (20.20% importance) – Investment levels significantly impact total contract values.
3. **AllocatedAreaFunding** (19.12% importance) – The area allocated for funding is a key factor.
4. **AllocatedAreaInvestment** (16.16% importance) – The area dedicated to investments also plays a role.
5. **CommercializationValue & IndustrializationValue** (~3.78% each) – Less impactful compared to other factors.

Model Performance

- **R² Score: 0.26**, indicating a moderate relationship. While the model captures some influence, additional factors may be at play.

This analysis confirms that **FundingValue** and **InvestmentValue** are the primary differentiators between high- and low-contributing financial institutions. 🚀

Would you like to refine this analysis further, such as testing a different model or adding categorical variables? [~]

Message Data Analyst



ChatGPT can make mistakes. Check important info.